

ClientFlow System Architecture & Guide

Rules, Boundaries, Roles & Operational Standard Operating Procedure (SOP)

1. Executive Overview & System Purpose

ClientFlow is an end-to-end MERN stack Onboarding & Client Management Platform designed to streamline the lifecycle between agencies, consulting teams, and onboarding clients. It enforces deterministic progression across client registration, proposal/contract execution, onboarding steps, asset exchange, and financial settlements.

2. How The System Works (Lifecycle Workflow)

Phase 1: Client Creation & Setup

Administrators or Team Members initiate client records in the system. A designated project is spawned with custom configurations, initial timezone setting, and an assigned Team Member.

Phase 2: Contract Execution & Compliance

A contract package is drafted and versioned with SHA hashes. The system dispatches digital execution requests. The client or authorized administrator executes the signature through the integrated e-signature module.

Phase 3: Financial Clearance (Invoices & Payments)

Prior to or alongside kickoff, milestone invoices are generated. Payments can be confirmed manually or via mock automated webhook triggers.

Phase 4: Multi-Step Interactive Onboarding

Clients progress step-by-step through customized onboarding workflows (Welcome, Information Collection, Credential Handover, Requirement Review). Each step validates required inputs before allowing completion.

Phase 5: Asset Exchange & Completion Review

Clients upload required brand identity assets and documentation via strict mime-type verified upload pipelines. Team Members review submitted onboarding instances and issue formal approvals or revision requests.

Role Matrix & Boundaries

Detailed Permissions, Operational Scopes & Security Fences

3. Role Matrix & Responsibilities

Role: Administrator (Admin)

Total operational governance and control over the platform, users, and audit records.

Core Responsibilities:

- **User Governance:** Create, suspend, and configure Admin, Team Member, and Client accounts.
- **System-Wide Monitoring:** Inspect financial health, audit logs, overall project delivery, and system statistics.
- **Contract & Invoice Oversight:** Issue invoices, draft contracts, override approval states, and handle exceptions.

Boundaries & Restrictions:

- **No Password Access:** Cannot retrieve cleartext user passwords; only resets can be triggered.
- **Immutable Audit Logs:** Cannot delete or alter append-only audit trail entries.

Role: Team Member

Dedicated project delivery specialists driving client onboarding and review processes.

Core Responsibilities:

- **Project Management:** Manage assigned client workspaces, track project milestones, and update statuses.
- **Onboarding Review:** Review submitted client onboarding documentation, approve steps, or request revisions.
- **Client Collaboration:** Monitor asset uploads, dispatch invoices, and maintain milestone communication.

Boundaries & Restrictions:

- **No User Administration:** Strictly prohibited from creating user accounts or editing global system settings.
- **Project Scope Restriction:** Restricted from modifying projects outside their assigned operational territory.

Role: Client User

External stakeholder interacting with assigned project workspaces and workflows.

Core Responsibilities:

- **Execution & Compliance:** Digitally execute contracts and fulfill invoicing obligations.
- **Onboarding Submissions:** Fill out required wizard questions, provide brand details, and upload assets.

Boundaries & Restrictions:

- **Strict Workspace Isolation:** Zero access to other client data, other projects, internal logs, or global invoices.
- **Read-Only Administrative Fields:** Cannot adjust template structures, change assigned team members, or alter financial totals.

Operational Rules & Best Practices

Security Policies, Step Protocols, and Data Handling Rules

4. Rules to Follow (System Operational Policies)

Authentication & Session Governance

- **HttpOnly Security:** All sessions rely on strict SameSite HttpOnly cookies. Never attempt to inject tokens into client localStorage.
- **Access Credentials:** Admin credentials (admin@clientflow.com) must be stored in secure vaults and rotated upon personnel changes.
- **Password Hygiene:** Passwords must satisfy complexity checks and bcrypt salted hashing.

Data Integrity & Audit Compliance

- **Append-Only Logging:** Every state transition (CREATE, UPDATE, ARCHIVE, SIGN, REVIEW) triggers an immutable audit log entry.
- **Idempotent Invoicing:** Invoice reference numbers are unique and must follow INV-{timestamp} or predefined sequential indexing.
- **Contract Hash Verification:** Every contract upload generates an immutable version hash preventing document repudiation.

Asset Handling & Validation Guidelines

- **Strict File Allowlist:** Only supported file types (PDF, PNG, JPG, DOCX, XLSX, ZIP) under 10MB are accepted by the pipeline.
- **UUID Sanitization:** Uploaded files are randomized using UUIDv4 filenames to completely eliminate directory traversal and collision risks.

5. Quick Reference: Seed Login Accounts

Role	Email Identifier	Default Password
Administrator	admin@clientflow.com	Admin@123
Team Member 1	team1@clientflow.com	password123
Team Member 2	team2@clientflow.com	password123
Client Stakeholder	client@clientflow.com	password123